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Multi-head Abductive Attention

Gary Nan Tie, Aug 25, 2025

Abstract

Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. We introduce an additional primitive to the vocabulary of formal query updating. The new construct updates a query in parallel, learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

Introduction

Multiple attention heads in transformers are used to enhance their expressive power and modeling capabilities. Rather than only computing attention once, a multi-head mechanism runs through the scaled dot-product attention multiple times in parallel. The independent attention outputs are simply concatenated and linearly transformed into the expected dimensions. This approach jointly models different types of dependencies and relationships from different representation subspaces at different positions. By learning multiple sets of attention parameters, the model becomes more robust and adaptable to different tasks and datasets. However, this does not take into account potentially different key-value pairing semantics. Attention for queries should be able to disambiguate different semantics at play and as well as synthesize their interactions, to give more meaningful answers. To remedy this, we propose initially updating a query using scaled kernel attention, then in parallel, learning multiple abductive fibration liftings to semantically refine similarity attention, yielding a final query update that is not only similarity grounded but also best adapted semantically.

Multi-head abductive fibration attention:

Queries q_i , $i \in [n]$, keys $k_j \in K$ and values $v_j \in V$, $j \in [m]$

Input query q_i , either raw or updated by some attention mechanism.

Define $q_i^0 \triangleq \sum_{j=1}^m \alpha_{ij} v_j$, $\alpha_{ij} = \text{softmax score}(q_i, k_j)$

where score is a similarity function like a reproducing kernel,

and let $j^0 \triangleq \arg \max_j \alpha_{ij}$

Let $p: K \rightarrow V$ with $p(k_j) = v_{j^0}$ and let $m(q_i) = q_i^0$,

learn σ_p from:

$$\begin{array}{ccc}
 Q \otimes K & \xrightarrow{m \otimes 1} & Q' \otimes K \\
 1 \otimes p \downarrow & \sigma_p \nearrow & \downarrow 1 \otimes p \\
 Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array}$$

That is learn $\sigma_p: Q \otimes V \rightarrow Q' \otimes K$ from I/O :

$$i) \quad \sigma_p \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} =_{\varepsilon} \begin{bmatrix} m(q_i) \\ k_j \end{bmatrix}$$

$$ii) \quad \pi_1 \sigma_p \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_{\varepsilon} m(q_i) \quad \left(\dagger \right)$$

$$iii) \quad p \pi_2 \sigma_p \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_{\varepsilon} v_j$$

Suppose we are given N key-value semantic pairings

$p_h: K_h \rightarrow V_h$ and we have learned using $(\dagger)$

$$\sigma_{p_h}: Q \otimes V_h \rightarrow Q' \otimes K_h, \quad h \in [N].$$

$$\text{Define } q_i^{1,h} \triangleq \pi_1 \sigma_{p_h} \begin{bmatrix} q_i^0 \\ v_{j^0} \end{bmatrix}, \quad h \in [N].$$

So the liftings σ_{p_h} of fibrations $1 \otimes p_h$

have refined q_i^0 into $q_i^{1,h}$, in parallel.

There is a head h and key j that maximizes

$$\left\{ \text{score}(q_i^{1,h}, k_j) \right\}_{h=1, j=1}^{N, m}$$

denote these by h^* and j^* .

$$\text{Let } q_i^1 \triangleq \begin{cases} q_i^{1,h^*}, & \text{if } \text{score}(q_i^{1,h^*}, k_{j^*}) > \text{score}(q_i^0, k_{j_0}) \\ q_i^0, & \text{otherwise.} \end{cases}$$

Query update process (multi-head):

$$\text{Input } q_i \mapsto \left. \begin{array}{c} \xrightarrow{\quad} q_i^{1,1} \\ \vdots \\ \xrightarrow{\quad} q_i^{1,N} \end{array} \right\} q_i^1 \text{ output.}$$

If $q_i^1 = q_i^{1,h^*}$, return value $P_{h^*}(k_{j^*})$,

otherwise return v_{j_0} .

—

This adds a new primitive to the formal query updating foundation for attention algorithm patterns.

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[1] 'Formal query updating to generate problem specific fibration abduction attention algorithms'

Gary Nan Tie, Aug 23, 2025

DOI: 10.13140/RG.2.2.33029.41446

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[2] 'FibChat - It makes stuff up, but with structure.'

Gary Nan Tie · July 2025

DOI: 10.13140/RG.2.2.10549.59364/2

FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[3] 'Fibrations explain all you need!

- Fibrations, Abduction, Attention

Gary Nan Tie, Mar 4, 2025

DOI: 10.13140/RG.2.2.35984.52488

To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism. Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.